

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\left(\phi_1(x), \begin{pmatrix} \phi_2(x) \\ \phi_3(x) \end{pmatrix}, \begin{pmatrix} \phi_4(x) \\ \phi_5(x) \end{pmatrix}\right) \rightarrow \left(c e^{\lambda x} \phi_1(x), U \begin{pmatrix} e^{-\lambda' x} \phi_2(x) \\ e^{-\lambda' x} \phi_3(x) \end{pmatrix}, V \begin{pmatrix} e^{\lambda'' x} \phi_4(x) \\ e^{\lambda'' x} \phi_5(x) \end{pmatrix}\right)$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial L}{\partial w_{ijkl}}(g_{ij}, g_{ij,k}, g_{ij,kl}) = \frac{\partial x^i}{\partial \tilde{x}^a} \frac{\partial x^j}{\partial \tilde{x}^b} \frac{\partial x^k}{\partial \tilde{x}^c} \frac{\partial x^l}{\partial \tilde{x}^d} \frac{\partial L}{\partial w_{abcd}}(\tilde{g}_{ij}, \tilde{g}_{ij,k}, \tilde{g}_{ij,kl}).$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$q^Q \prod_{k=1}^{Q-1} \frac{qz_i - z_k}{z_i - qz_k} = \frac{\left(z_i - i\tau\kappa\lambda^{-\frac{1}{2}}q^{\frac{1}{2}}\right)\left(z_i + i\kappa\lambda^{\frac{1}{2}}q^{\frac{1}{2}}\right)}{\left(q^{\frac{1}{2}}z_i + i\tau\kappa\lambda^{-\frac{1}{2}}\right)\left(q^{\frac{1}{2}}z_i - i\kappa\lambda^{\frac{1}{2}}\right)}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\overrightarrow{\not{D}}_R \equiv e_a^\mu \gamma^a \left(\vec{\partial}_\mu + \frac{1}{2} \omega_\mu^{cd} \sigma_{cd} + i g' B_\mu \right); \quad \overleftarrow{\not{D}}_R \equiv \left(\overleftarrow{\partial}_\mu - \frac{1}{2} \omega_\mu^{cd} \sigma_{cd} - i g' B_\mu \right) \gamma^a e_a^\mu;$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\frac{d^2 y}{d\tilde{\lambda}^2} - \frac{1}{2} \frac{\mathcal{W}'}{\mathcal{W}} \left(\frac{dy}{d\tilde{\lambda}} \right)^2 - \frac{1}{2} \left[\mathcal{W} - \left(\frac{dy}{d\tilde{\lambda}} \right)^2 \right] \mathcal{W}^{-1} \partial_y (\mathcal{W} g_{\mu\nu}) \frac{dx^\mu}{d\tilde{\lambda}} \frac{dx^\nu}{d\tilde{\lambda}} = 0.$$